Certified Identifiability at Spectral Degeneracies

Sharp Moduli, Minimal Probes, and What Finite Data Can Say About Holonomy

Matěj Rada
MatejRada@email.cz
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Abstract

Real symmetric families have eigenvalue collisions, and the eigenlines around such a collision carry a sign holonomy. We determine exactly what calibrated measurements can and cannot certify about these lines and holonomies, under adversarially bounded errors. The results are as follows.

(i) *Pointwise moduli.* The fixed-spectrum pairwise modulus is $\min(\varepsilon/g, 1)$. A general isolated eigenline obeys $\|P_A - P_B\| \le \varepsilon/(g-\varepsilon)$. The extremal eigenline obeys the *sharp* bound $\sin(\frac{1}{2}\arcsin(2\varepsilon/g))$, which exceeds ε/g at finite error.

(ii) *Coarse data and minimal probes.* Eigenvalues together with coordinate intensities can collide at positive gap. Exactly $n-1$ additional calibrated intensity analyzers are necessary and sufficient to recover every real eigenline.

(iii) *Holonomy from loops.* Sign holonomy is recovered from noisy node lines iff the largest cell budget ℓ_i satisfies $\ell_i + 2\alpha < \pi/2$, or the total budget B is below π. The uniform-mesh threshold is $\max(\pi/N, \pi/2 - 2\alpha)$, sharp for *every* estimator, with

$$N_{\min} = \lfloor B/(\pi/2 - 2\alpha) \rfloor + 1.$$

A data-dependent certificate is strictly stronger than blanket abstention.

(iv) *Clusters.* The orientation part $\det H$ of a rank-m cluster holonomy obeys the same thresholds. The continuous part of the holonomy is *never* determined by finitely many node planes; it is set-valued, with an isoperimetric radius.

(v) *Statistics.* With Gaussian angular noise, the sample size needed per node is pinned between $2z_p^2\sigma^2/\Delta^2$ and $8\log(2N/p)\,\sigma^2/\Delta^2$, where $\Delta = \pi/2 - s$.

(vi) *Adaptivity.* Without a declared path budget no finite adaptive schedule can certify parity.

(vii) *Apparatus.* In a simulated pair of coupled oscillators encircling the degeneracy, the readout sign flips, the adiabatic error decreases with duration, and a non-encircling control does not flip.

All proofs are self-contained and the batteries are deterministic. All data are synthetic.

The final section states what this closes for Resolution Geometry. It is the first complete instance of RG's information-loss classification: sharp moduli, minimal probes, typed abstention, and a proof that the continuous part of a holonomy is a hidden value that no finite protocol fixes.

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1 Contracts

- **Matrices.** Matrices are real symmetric $n \times n$ ($n \geq 2$) in a calibrated ambient basis. For a simple tracked eigenvalue, the task is the projector $P = uu^\top$, with metric $\|P - Q\| = \sin\theta$. A signed eigenvector is gauge-dependent and is not a task.
- **Loops.** A real line loop $P : [0, L] \to \mathbb{RP}^{n-1}$ is closed and absolutely continuous. A continuous unit lift satisfies $u(L) = h\,u(0)$, and the sign holonomy $h \in \{\pm 1\}$ is the task.
- **Measurements and budgets.** Node measurements are lines within projective angle α of the truth; errors are adversarial, with no independence assumed. Each cell carries a certified path budget ℓ_i, and $B = \sum \ell_i$.
- **Assumptions.** Closure, basis calibration, branch association and the declared bounds are assumptions, not conclusions drawn from samples.

2 Pointwise identifiability

Theorem 2.1 (Fixed-spectrum modulus)**.** *For* $A(P) = \lambda_- I + gP$, $\sup\{\|P - Q\| : \|A(P) - A(Q)\| \leq \varepsilon\} = \min(\varepsilon/g, 1)$.

Proof. $A(P) - A(Q) = g(P - Q)$, and rotating one line attains every distance in $[0, 1]$. A separated spectator embeds the example in every n. □

Theorem 2.2 (Isolated eigenline)**.** *Let* λ *be a simple eigenvalue with gap* g*, and let* $\|E\| \leq \varepsilon < g/2$*. Then* $\|P_A - P_{A+E}\| \leq \varepsilon/(g - \varepsilon)$.

Proof. By min-max, the tracked eigenvalue shifts by at most ε. Project $(A - \mu)v = -Ev$ onto $\ker P_A$. The inverse of $A - \mu$ restricted there has norm at most $1/(g - \varepsilon)$. □

Theorem 2.3 (Sharp extremal modulus)**.** *For the top eigenline with gap* g *and* $\|E\| \leq \varepsilon < g/2$*, the exact worst-case error is*

$$\sin\left(\tfrac{1}{2}\arcsin(2\varepsilon/g)\right).$$

Proof. Upper bound. Write $v = \cos\theta\,u + \sin\theta\,w$ and $z = -\sin\theta\,u + \cos\theta\,w$. Then $\varepsilon \geq |z^\top E v| \geq g\sin\theta\cos\theta$. Continuation along $A + tE$ keeps the gap open, so $\theta < \pi/4$.

Attainment. Take $A = \frac{g}{2}\mathrm{diag}(1, -1)$ and $B = \frac{g}{2}c\begin{pmatrix} c & s \\ s & -c \end{pmatrix}$ with $c = \cos 2\theta$, $s = \sin 2\theta$. □

At $\varepsilon/g = 0.3$ the attained error is $0.3162 > 0.3$, so ε/g alone is not a uniform finite-error bound.

Theorem 2.4 (Coarse collision and minimal analyzers)**.** *Eigenvalues together with* $\mathrm{diag}(P)$ *do not identify* P*: the lines* $(\cos a, \pm\sin a)$ *have equal coarse data at distance* $|\sin 2a|$*, at positive gap. Given all* n *coordinate intensities, exactly* $n-1$ *additional fixed real analyzer intensities* $|a_j^\top u|^2$ *are necessary and sufficient to recover every real line.*

Proof. Sufficiency. Choose the $2n-1$ analyzers full spark: every n of them span. If u and v give equal intensities, each analyzer is orthogonal to $u-v$ or to $u+v$. One of these sets contains at least n analyzers, which span, so $v = \pm u$.

Necessity. With at most $2n-2$ analyzers, split them into two sets of size at most $n-1$, and take nonzero x, y orthogonal to the respective spans. Because the coordinate axes are among the analyzers, $x \perp y$. Then $x+y$ and $x-y$ form an alias. □

For example, one extra analyzer suffices in 2D, namely $(1,1)/\sqrt{2}$, which gives the exact repaired modulus $\min(\varepsilon, 1)$. Two suffice in 3D, namely $(1,1,1)/\sqrt{3}$ and $(1,2,3)/\sqrt{14}$.

3 Holonomy from loops

Theorem 3.1 (Sufficiency; global exception)**.** *If* $\ell_i + 2\alpha < \pi/2$ *on every cell, then*

$$\textstyle\prod_i \mathrm{sign}(v_i^\top v_{i+1}) = h$$

for arbitrary representatives v_i *of the measured lines.*

If $B < \pi$*, then* $h = +1$ *for every admissible loop, and no data are needed.*

Proof. Lift each cell continuously. Measured representatives within α of the lifts are separated by at most $\alpha + \ell_i + \alpha < \pi/2$, so each lifted overlap is positive. Node signs cancel in the product.

A negative loop must travel from u to $-u$, which needs spherical length at least π. □

Theorem 3.2 (Exact threshold, every estimator)**.** *Opposite-parity aliases sharing a common noisy dataset exist if and only if*

$$B \geq \pi \quad \textit{and} \quad \max_i \ell_i + 2\alpha \geq \pi/2.$$

For a uniform mesh with $N \geq 2$ *cells and per-cell budget* s*, this reads* $Ns \geq \pi$ *and* $s + 2\alpha \geq \pi/2$*. The threshold in* s *is* $\max(\pi/N, \pi/2 - 2\alpha)$*, and when* $B \geq \pi$ *and* $\alpha < \pi/4$*,*

$$N_{\min} = \lfloor B/(\pi/2 - 2\alpha) \rfloor + 1.$$

Proof. Necessity follows from Theorem 3.1.

For sufficiency, put the largest budget s first and let $S = B - s$ be the remaining budget. The construction depends on s.

- *If* $s < \pi/2$*:* start the two lifts at $a_0 = 0$ and $b_0 = r = \frac{\pi - 2s}{2}$, with first increments $(-s, +s)$. On the remaining cells use increments $s\ell_i/S$ and $(\pi - s)\ell_i/S$. The node lines stay within r of each other modulo π, so midpoint lines lie within $r/2 \leq \alpha$ of both.
- *If* $\pi/2 \leq s < \pi$*:* start both lifts at 0, with first increments $(s - \pi, s)$ and equal increments afterwards. The node lines agree exactly.
- *If* $s \geq \pi$*:* one loop makes a half-turn inside the first cell and the other stays constant.

In every case the endpoints differ by a half-turn, so the parities are opposite. Placing nodes at equal increments of the budget attains $N_{\min}$; with fewer nodes, some cell has $\ell_i \geq B/N$. □

Theorem 3.3 (Data-dependent certificate)**.** *Let $\theta_i < \pi/2$ be the observed angle between neighbouring node lines. If $\ell_i + 2\alpha + \theta_i < \pi$ on every cell, the overlap product equals h.*

Proof. An opposite-sign transition would need a lifted path of length at least $\pi - \theta_i - 2\alpha$, which exceeds the budget. □

Blanket abstention is therefore conservative. Minimax sharpness says only that some ambiguous dataset exists in the regime, not that every abstained dataset is ambiguous.

4 Clusters: discrete versus continuous holonomy

Theorem 4.1 (Orientation holonomy of rank-m clusters)**.** *For a loop of rank-m projectors with principal-angle budgets ℓ_i and node errors α, consider the product $\prod_i \operatorname{sign}\det(F_i^\top F_{i+1})$ computed from arbitrary orthonormal frames F_i of the measured planes. It equals the orientation holonomy $\det H$ under the conditions of Theorem 3.1, and the threshold of Theorem 3.2 is exact.*

Proof. Gauge. Changing the frame at node i multiplies two adjacent determinants by $\det g_i$, which cancels.

Positivity. Along a lifted cell all principal angles stay below $\pi/2$, so the overlap determinant never vanishes and stays positive by continuity from 1.

Sharpness. Embed the rank-one aliases as the span of a moving line and a fixed $(m-1)$-plane. □

Battery R1: 1500+ random rank-m loops (with $n = 3, \ldots, 6$), arbitrary node frames, no failures.

Theorem 4.2 (Continuous holonomy is not finitely identifiable)**.** *Let H_{can} be the holonomy of the canonical (tautological) connection. Its continuous part is not determined by any finite set of node planes when the cell budgets are positive. For planes in $\mathbb{R}^3$, the canonical holonomy angle equals the solid angle enclosed by the normal loop (Gauss–Bonnet).*

Two admissible paths between the same nodes therefore differ by their enclosed-area difference. The finite-sample task is set-valued, of radius at most $\sum_i A(\ell_i + c_i)$, where c_i is the chord between neighbouring nodes and $A(L) = 2\pi - \sqrt{4\pi^2 - L^2}$ is the spherical isoperimetric bound [5].

Witness (R2). Take six node planes with normals on the circle at colatitude 0.5. The geodesic polygon gives holonomy 0.6555, an outward-bulged path 0.3788, and an inward-bulged path 0.9322. The differences are 0.277, within the bound $6A = 0.546$.

The discrete part of the holonomy ($\det H$) is finitely certifiable. The continuous part is a hidden value that requires a path or area budget.

5 Composition and statistics

Composition. A measured matrix with gap $\hat{g} > 2\delta$ gives $\alpha \le \arcsin(\delta/(\hat{g} - \delta))$. For a C^1 family with $\|A'\| \le M$ and global gap $\ge g_{\min}$, the angular speed satisfies $\|P'\| \le M/g_{\min}$. Moreover $g_{\min} \ge \min_i \hat{g}_i - 2\delta - Mh$. Chaining these with Theorems 3.1–3.2 is deterministic.

Confidence transfer. If the calibration event holds with probability at least $1 - p$, a sound certificate errs with probability at most p.

Theorem 5.1 (Statistical rate)**.** *Suppose each node angle is observed with i.i.d. Gaussian error of variance σ^2/m. Uniform mesh, $s \in [\pi/4, \pi/2)$, $N \ge 4$, $\Delta = \pi/2 - s$.*

- Lower bound. *An explicit alias pair differs as lines at only two nodes, each by Δ. Its Bayes error is therefore $\Phi(-\Delta\sqrt{m/2}/\sigma)$ (wrapping can only increase it).*

- Upper bound. *The certificate with* $\alpha = \sigma\sqrt{2\log(2N/p)/m} < \Delta/2$ *errs with probability at most* p.

Hence

$$2z_p^2\sigma^2/\Delta^2 \;\le\; m^* \;\le\; 8\log(2N/p)\,\sigma^2/\Delta^2.$$

Witness (R3). The alias construction was checked for $N \in \{4, 6, 12, 40\}$ and four values of s: budgets respected, opposite parities, and exactly two differing nodes. A Monte Carlo Bayes error of 0.0950 matches the formula's 0.0949.

No-Go 5.2 (Unknown speed). If no path budget is declared, then for any finite and possibly adaptive sampling schedule there exist two loops that agree on every sample and have opposite parity: insert a half-turn inside an unsampled interval. Optimal adaptive design therefore requires a declared budget, and with a known budget density the optimal fixed design is given by Theorem 3.2.

6 Synthetic apparatus

Take two coupled oscillators $\ddot{x} + K(t)x = 0$ with $K = \begin{pmatrix} k_0 + d & c \\ c & k_0 - d \end{pmatrix}$, where (d, c) traverses a circle of radius 0.4 in time T, and $k_0 = 2$. Mode 1 is excited. The readout is $\mathrm{Re}(z\,e^{i\int\omega\,dt})$, where z is the complex amplitude projected onto the initial mode and the dynamic phase comes from the known model.

T	50	100	200	400	800
encircling (truth -1)	-0.721	-0.919	-0.985	-0.996	-0.999
control (truth $+1$)	0.925	0.979	0.9986	0.9998	0.99996

The adiabatic error decreases by roughly a factor of 4 per doubling of T. This is a simulation: the dynamic phase is assumed known, and no apparatus noise or calibration study is included.

7 What this closes for Resolution Geometry

1. **The information-loss programme has its first complete instance.** For one object class it provides:
   - exact pointwise moduli;
   - an exact minimal-probe theorem with lower bound ($n - 1$ analyzers);
   - exact loop thresholds for every estimator;
   - a typed abstention that is provably conservative;
   - a statistical rate.

   This is RG's rule (descend, or return the compatible set, or abstain with justification) realized with sharpness on both sides.

2. **Forced monodromy (L122) becomes certifiable from finite noisy data,** with the exact condition $\max \ell_i + 2\alpha < \pi/2$ or $B < \pi$.

3. **The mesh-closure coverage condition becomes sharp.** The heuristic "step below π" of the mesh paper is replaced by the exact $\max(\pi/N, \pi/2 - 2\alpha)$ for line holonomy.

4. **Hidden rooms are typed precisely.** The discrete (orientation) holonomy is finitely certifiable. The continuous holonomy is a hidden value that no finite protocol fixes; it is set-valued with an isoperimetric radius. This sharpens the functional link moduli of L130.

5. **Abstention semantics.** A data-dependent certificate certifies datasets outside the minimax region, so an abstention is a property of the dataset, not of the regime.

Not closed.

- extension classification across *changing* rank or stabilizer type (T1 in general);
- sharp analyzer conditioning for $n \geq 3$;
- interior-eigenline sharp constants;
- statistical lower bounds beyond the two-point construction;
- real apparatus data, dynamic-phase calibration and finite-rate error bounds;
- a literature novelty audit, notably against perturbation theory, phase retrieval and topological sampling.

8 Ledger

Result	Status	Evidence
Fixed-spectrum, isolated and sharp extremal moduli	proved	core battery: 60 attainment, 1800 perturbation, 800 interior checks
Coarse collision; $n-1$ analyzers	proved	219 reconstructions; 475 exact minors; lower-bound aliases
Sufficiency, global exception	proved	1200 nonplanar noisy certificates
Exact threshold, $N_{\min}$, nonuniform iff	proved	693 alias embeddings; 133 exact rational; 1092 nonuniform aliases
Data-dependent certificate	proved	core battery
Rank-m orientation holonomy	proved	R1
Continuous holonomy not finitely identifiable	proved	R2
Statistical rate	proved (two-point)	R3 (Monte Carlo 0.0950 against 0.0949)
Unknown-speed no-go	proved	construction
Coupled-oscillator readout	simulated	R4
Changing-rank extensions; real data	open	—

9 Reproduction and limits

Reproduction.

- **Core package** (all reran cleanly):
  - `rg_spectral_lock.py`
  - `rg_spectral_lock_validate.py`
  - `rg_nonuniform_alias_validate.py`
  - `rg_spectral_additional_checks.py`
- **Closure:** `rg_spectral_closure.py` (R1–R4; byte-identical reruns).

Limits. All data are synthetic. Floating-point checks are not interval-certified. The proofs are self-contained but not machine-checked. The battery is not blind, and no external peer review has taken place.

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